

Pamidronate

Pamidronate is used to treat and prevent problems caused by the effect of cancer on the bones.

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What is pamidronate?

Pamidronate belongs to a group of drugs called [bisphosphonates](#). It can be used to treat:

- high levels of calcium in the blood caused by cancer that has spread to the bones ([secondary bone cancer](#))
- bone weakness or pain caused by [myeloma](#) or by [breast cancer](#) that has spread to the bones.

Pamidronate can be given with [other cancer treatments](#).

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

The effect of cancer on the bones

Bone cells constantly renew themselves. This helps bones keep their strength and shape.

There are 2 types of cell inside the bones. These are called:

- osteoclasts, which break down and remove old bone
- osteoblasts, which build new bone.

As we get older, osteoclasts begin to break down more bone than osteoblasts make to replace it. This means our bones slowly become thinner (less dense). In some people, too much of the bone density is lost and this increases the risk of osteoporosis and bone fractures.

Some [secondary bone cancers](#) make a substance that causes osteoclasts to destroy more bone than can be replaced. The affected bone becomes weak and painful and can break more easily.

Bones contain calcium, which gives them strength. A bone affected by cancer may cause calcium from the bones to be released into the blood. A raised level of calcium in the blood is called hypercalcaemia. This may cause symptoms including:

- feeling sick (nausea)
- being sick (vomiting)
- feeling thirsty
- needing to pee (pass urine) more often
- constipation
- tiredness
- irritability
- confusion.

How pamidronate works

Pamidronate reduces the activity of osteoclasts. This can help to reduce pain and strengthen the bone.

Pamidronate also reduces the amount of calcium lost from the bones. This helps calcium levels in the blood return to normal.

How pamidronate is given

You will be given pamidronate in the outpatient department at the hospital. You will have blood tests before starting treatment and during treatment. These check several things, including:

- how your kidneys are working

- your calcium levels.

Your urine may also be checked.

Pamidronate is given by a drip (infusion) into a vein. The infusion may be given over between 90 minutes and 4 hours. Your doctor or nurse will tell you how long your treatment will take. It is usually given every 3 to 4 weeks.

How long pamidronate is given for

If you are having pamidronate to reduce pain or strengthen your bones, you may continue it for as long as it helps manage the symptoms.

If pamidronate is given to lower calcium levels, it is usually given as a single dose. Or you may be given several infusions over 2 to 4 days.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Dental check-up before treatment

Rarely, this treatment can affect the jawbone. You can read more about this in the side effects section below.

Keeping to a regular oral hygiene routine keeps your mouth and teeth healthy. This reduces your risk of jaw problems. You usually need a dental check-up to check for any problems before you start treatment.

If you need urgent treatment to lower high calcium levels in your blood, you do not usually need to have a dental check-up first.

Side effects while treatment is being given

Some people may have side effects while they are having a cancer drug or shortly after they have it:

Allergic reaction

Rarely, some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Pain along the vein

This treatment can cause pain:

- at the place where the drip (infusion) is given
- along the vein.

If you feel pain, tell your nurse straight away. They can check the site. They may give the drug more slowly or flush it through with more fluid to reduce pain.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Flu-like symptoms

Some people have flu-like symptoms, such as:

- a high temperature
- chills
- pains in your muscles or joints.

Tell your doctor if these effects are causing problems. It may help to take mild painkillers, such as paracetamol. Your doctor, nurse or pharmacist can give you advice.

Numbness or tingling

You may notice numbness or tingling around the mouth, or in the fingers and toes. Contact your doctor or the hospital straight away if you have these symptoms. This may be caused by low levels of calcium in your blood. You will have regular blood tests to check your calcium levels.

Your doctor may ask you to take calcium and vitamin D supplements, unless you are having this treatment to lower the levels of calcium in your blood. They will tell you if you need any supplements. This will depend on blood test results.

Muscle weakness

If you have low levels of phosphate in your blood, you may have muscle weakness. You may also feel confused. You will have regular blood tests to check your phosphate levels. Contact the hospital on the 24-hour number you have given if you have any of these symptoms.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Increased bone pain

Sometimes pain in the affected bone gets worse for a short time when you start taking this treatment. If this happens, your doctor can prescribe painkillers to help.

Feeling sick (nausea)

You may [feel sick](#) or be sick (vomit) while having this treatment. This is usually mild. Your doctor can prescribe anti-sickness drugs to help prevent or control sickness.

Take the drugs exactly as your doctor, nurse or pharmacist tells you to. It is easier to prevent sickness than treat it after it has started.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Diarrhoea or constipation

Tell your doctor if you have [diarrhoea or constipation](#). They can give you medicine to help with this. Try to drink at least 2 litres (3½ pints) of fluids each day.

Red or sore eyes

This treatment may make your eyes feel sore, red and itchy (conjunctivitis). If you have red or sore eyes, your doctor can prescribe eye drops to help treat this. Your pharmacist can tell you how to use the eye drops.

Skin changes

This treatment can cause a rash. Tell your doctor if this happens. They can give you advice and may prescribe creams or medicines to help. Any changes to your skin are usually temporary and improve when treatment finishes.

Low number of white blood cells

This treatment can reduce the number of white cells in your blood. These cells fight infection. If you have any signs of an [infection](#), tell your doctor, nurse or pharmacist. These signs include:

- a cough
- a sore throat
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Blood pressure changes

This treatment can affect your blood pressure. It can cause high blood pressure and sometimes low blood pressure. Tell your doctor if you:

- have problems with your blood pressure
- have headaches
- feel dizzy.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Effects on the kidneys

This treatment can affect how the kidneys work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Jaw problems

Rarely, this treatment can affect the jawbone. Healthy bone in the jaw becomes damaged and dies. This is called osteonecrosis of the jaw (ONJ). It can cause:

- pain
- loosening of the teeth
- problems with the way the gums heal.

The risk of jaw problems is higher:

- after some types of dental treatment
- if you smoke
- if you have gum disease
- if you wear dentures that do not fit well.

It is important to avoid having any dental treatment that could affect your jawbone when you are having bisphosphonates. This includes having a tooth or root removed, or dental implants put in. You can still have fillings, gum treatments or a scale and polish. Your healthcare team can tell you more.

To reduce your risk of developing jaw problems, your doctor will advise you to:

- have a full dental check-up before starting treatment
- have good oral hygiene by looking after your teeth and gums during treatment – ask your dentist for advice
- have regular dental check-ups
- tell your dentist you are taking a bisphosphonate before having any dental treatment.

Tell your doctor and dentist straight away if you develop:

- pain
- swelling or redness in your gums
- numbness or heaviness in your jaw
- loose teeth.

Pain in the thigh, hip or groin

Rarely, people having this treatment develop a break (fracture) in their thigh bone without an obvious cause. Sometimes both thigh bones are affected.

If you have any thigh, hip or groin pain, tell your doctor. Let them know that you are taking bisphosphonates. They can arrange tests to check the thigh bones for any signs of bone weakness or fracture.

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

Bone problems in the ear

Very rarely, the small bones in the outer ear may be affected by this treatment. Always tell your doctor if you have any:

- ear pain
- discharge from your ear
- ear infections.

You should also let them know if you notice any other changes in your ears or hearing.

Risk of having a fit (seizure)

You may have a higher risk of having a fit (seizure). But this is rare. This may be caused by high levels of calcium from the cancer and from having treatment. If this happens, call 999 straight away.

Other important information

Going into hospital

If you are admitted to hospital for a reason not related to the cancer, it is important to tell the doctors and nurses that you are having treatment with bisphosphonates. You should tell them the name of your cancer specialist. This is so they can talk to them for advice.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements

- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Pregnancy

Ask your doctor or pharmacist for advice before taking this medicine if you:

- are pregnant
- think you may be pregnant
- are planning to have a baby.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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! This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

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We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

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